

Virtual interface wiring

Class-based testbenches need a bridge to the DUT pins

The wiring checklist

- Six steps, in order
- Declare the interface with clock and bus signals
- Instantiate it in the testbench module with clock connected
- Connect DUT ports to interface signals or a modport
- Declare a virtual interface handle inside the driver or monitor class
- Assign the handle , drv dot vif equals bif, or later a config database set and get

Browser lab

Digital Design and Verification Platform

HomeTools

Home / Tools / Virtual interface wiring

INTERACTIVE TOOL

Virtual interface wiring

Sketch how a class TB reaches DUT pins: **interface** → instance → DUT ports → virtual handle in a driver. Starter: UART-ish bus_if with three steps already wired.

Starter example: UART uart_if with declare, instance, and DUT connect already checked. Add virtual + assign.

Load starter example

Challenges 0 / 22 cleared

Quiz: why virtual: Classes use a virtual interface because...

☐ class objects cannot connect to module ports like a DUT instance

☐ synthesis requires virtual in every RTL module

☐ VCD dump only works with virtual

☐ objections need a virtual clock

Show hint

Check

Next

Idle

1

2

3

4

5

6

7

8

9

10

11

12

13

14

15

16

17

18

19

20

21

22

Core ideas

Interface

Bundles related DUT signals in one type.

Instance

Lives in the TB module next to the DUT.

virtual

Class handle that points at an IF instance.

Assign

Wire the handle (or ConfigDB set/get).

Browser lab starter

learn_sv_tb — vif wiring

Real SystemVerilog practice

- In the real SystemVerilog track, open this module's examples prompts
- Restate the wiring chain in one sentence
- Sketch the UART pattern below on paper and number the six checklist steps next to each
- Optional stretch
- ``systemverilog // 1-2

Pitfalls to watch

- Do not drive DUT ports directly from a class when an interface is the intended boundary
- Do not forget to assign the virtual handle before calling drive tasks
- Do not mix up virtual interface with a plain interface instance
- And remember

Your turn

- Complete the checklist for at least one track , preferably both
- In the browser, load the UART starter, check all six wiring steps
- On paper, draw the signal path from class through virtual handle to DUT pin
- When you are ready